

Project 3 – CLI Utilities with Files & Directories

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Background

We will work with files and directories in the Fedora server VM.

Objectives

1. Start Fedora server virtual machine.
2. Create a directory tree.
3. Move and rename files.
4. Remove files and directories.
5. Document steps for inclusion in a project report to be submitted for review and grading in Blackboard.

Prerequisites

1. A Windows or MacOS device with a functioning hypervisor (e.g., VMware Workstation Pro or VMware Fusion Pro).

Creating a directory tree and files

Note – the commands are written in brackets for you to be able to distinguish commands from the other text. When you are using the commands in Linux do not copy the [] or ().

Setup

1. Login using your non-root account. [**Login van**]
2. Use [**pwd**], check what is your *current directory* path /home/van . If it is not your home directory, then use [**cd ~**] to *go back*. Check with [**pwd**] again.

Prepare to back up your directory tree that is to be created

3. Create a new directory called *dirbak* under /tmp by entering [**mkdir /tmp/dirbak**].
4. In previous step, make sure you create the directory starting with a “/”, which is **/tmp/dirbak**, not **tmp/dirbak**. If you are unsure about this, check with instructor.

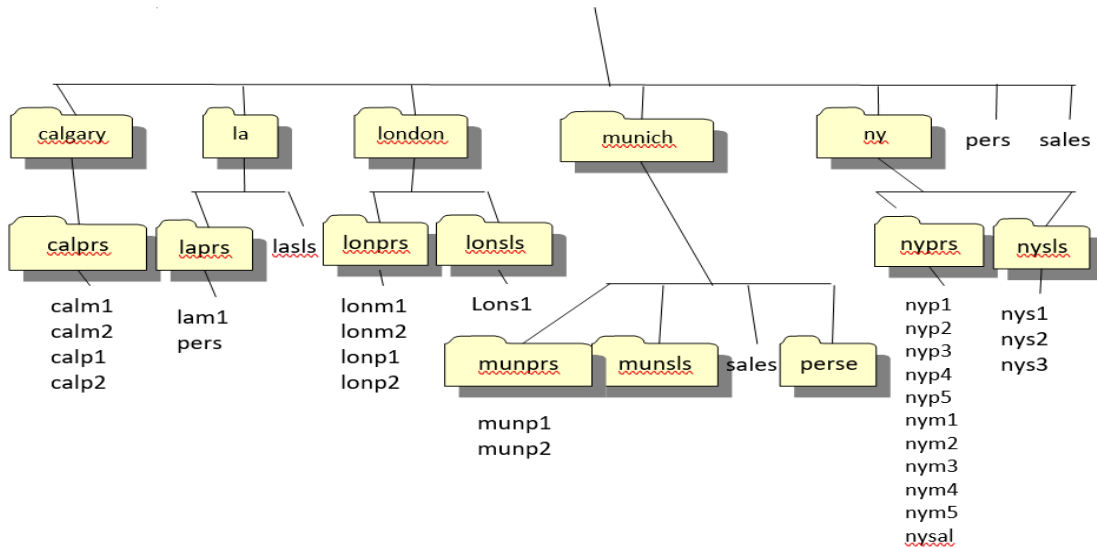
(For your information, /tmp directory is writable to every user. We will create a backup of your directory tree there.)



5. Create a directory tree with files:

Go to your home directory by entering `[cd ~]`. Create directories and files as shown below:

Your currently directory should be: `/home/your_username`



8. Verify the copy is successful by going to `/tmp/dirbak/` and check that the directories and files are there.

Moving and Renaming Files

1. Change to the `munich` subdirectory under your home directory.
2. List the files and subdirectories in `~/munich` by entering `[ls -F]`

Check the man page of `ls`, what is `-F` for?

`_ - F _` it means *Right* `list the contents of your current working directory`.

3. Enter `[mv sales munsls]`
4. Enter `[ls -F]` to list the contents of your current working directory, `~/munich`.



Notice that the sales file is no longer listed.

5. Change to the ~/munich/munsls directory and verify that sales have been moved here.

6. Enter `[mv sales munsls]`, and then do a directory listing.

Compare what happened with the results of step 3. What is the difference between the two?

`[mv sales munsls]` commands that are in step 3 and step 6?

7. Now, let's try moving a file to your current working directory. Change to your ~/munich directory, and then enter `[mv munsls/munsls sales]`. This will move the file you've been working with and rename it back to sales.

Check that the file is no longer in munich/munsls and that *sales* is in your ~/munich directory.

Removing Files and Directories

1. Change to the ~/munich directory, and then display a recursive directory listing `[ls -R]`. There are three directories and one file in the munich directory. The mnprs directory contains files and the munsls and pers directories are empty.

2. Enter `[rmdir munsls]`

Were you able to delete this directory? Why or why not? **Yes**

3. Enter `[rmdir munprs]`

Were you able to delete this directory? Why or why not?



4. Enter `[rm -ri munprs]` to interactively delete the munprs directory and all subdirectories and files in it. Respond with y at each prompt.

Verify that the directory and all its contents were removed `[ls -R ~/munich]`.

5. **Logout.**

Submission

Use this document (or one of your choosing) to submit your screen shots and any other relevant data. Ensure enough number of screen shots for the instructor to understand all the steps. At a minimum, you should show the commands run after logging in as root. Please ensure your submission includes your name, class, date, and reference to this lab.

